8	**(a)** <u>nuclei</u> having same number of protons/proton (atomic) number	B1	
	different numbers of neutrons/neutron number	B1	[2]
	(allow second mark for nucleons/nucleon number/mass number/atomic mass if made clear that same number of protons/proton number)		
	(b) probability of decay per unit time is the decay constant	C1	
	$\lambda = \ln 2 / t_{½}$		
	$= 0.693 / (52 \times 24 \times 3600)$	C1	
	$= 1.54 \times 10^{-7}\ s^{-1}$	A1	[3]
	(c) **(i)** $A = A_0 \exp(-\lambda t)$		
	$7.4 \times 10^6 = A_0 \exp(-1.54 \times 10^{-7} \times 21 \times 24 \times 3600)$	C1	
	$A_0 = 9.8 \times 10^6$ Bq	A1	[2]
	(*alternative method uses 21 days as 0.404 half-lives*)		
	(ii) $A = \lambda N$ and mass $= N \times 89 / N_A$	C1	
	mass $= (9.8 \times 10^6 \times 89) / (1.54 \times 10^{-7} \times 6.02 \times 10^{23})$		
	$= 9.4 \times 10^{-9}$ g	A1	[2]

Section B

9 (a) e.g. infinite input impedance/resistance